



Pre-Reads

14-Day AI Accelerator Program

Pre-Read III

Session Name:

AI Automations - Introduction to Automation with n8n

A Non-Technical Introduction to AI-Powered Automation

Your Essential Pre-Read for Mastering n8n: A Non-Technical Introduction to AI-Powered Automation



I. What is n8n?

n8n (pronounced *en-eight-en*) is a **visual workflow automation tool**. Imagine it as a digital

assembly line where you connect different software applications (apps), services, and data sources. Instead of writing complex code, you use simple drag-and-drop blocks called **Nodes** to create a sequence of actions.

II. Why is n8n a Game-Changer for AI Learners?

For those learning about Artificial Intelligence, n8n is incredibly powerful because it bridges the gap between AI models and everyday tools. You can easily:

- **Connect to AI Services:** Integrate models like OpenAI (ChatGPT), Cohere, or AI21 Studio into your workflows.
- **Automate AI-driven Tasks:**
 - Automatically classify incoming emails or customer feedback.
 - Generate summaries of articles or reports.
 - Extract specific information from large texts.
 - Translate languages.
- **Build Custom AI Agents:** Create systems that react to new data, make decisions using AI, and then take actions in other applications.

Essentially, n8n allows you to *apply* your AI knowledge in practical, automated ways without getting bogged down in complex programming.

III. Core Concepts: The Building Blocks of n8n

Understanding these terms will be key:

- **Workflow:** The entire automated process you design, from start to finish. It's the complete sequence of tasks.
- **Node:** A single step or task within your workflow. Examples: "Read from Google Sheets," "Summarise with OpenAI," "Send Email."
- **Trigger Node:** The special node that starts your workflow. This could be:
 - **Scheduled:** Runs at a specific time (e.g., every Monday at 9 AM).

- **Webhook:** Starts when it receives data from an external app.
- **Manual:** You click a button to start it.
- **Event-based:** Starts when something happens in another app (e.g., a new row in Airtable).
- **Execution:** A single run of your workflow. If your workflow runs 10 times, that's 10 executions.
- **Connection (Credentials):** The secure, authenticated link between n8n and an external application (like your Google account, OpenAI API key, etc.). This tells n8n it has permission to access that app on your behalf.
- **Integration:** The built-in support n8n offers for a specific tool or platform (e.g., n8n has integrations for Google Sheets, Slack, OpenAI, Notion, etc.).
- **Action Nodes** – Send data or perform tasks
- **Data Transformation Nodes** – Reformat or filter your data
- **Logic Nodes** – Make decisions (e.g., IF, SWITCH)
- **Custom Sub-Workflows** – Reusable tools called by other workflows

IV. How a Workflow Works: The Flow of Data

Workflows in n8n operate on a simple principle: **data flows from one node to the next.**

1. **Start with a Trigger:** Let's say a "Schedule Trigger" runs your workflow every morning.
2. **Add an Action Node:** This trigger passes information (like the current time) to the next node. Perhaps an "HTTP Request Node" to fetch top news headlines.
3. **Process with Another Node:** The headlines (data) are passed to an "OpenAI Node" to generate a summary for each.
4. **Final Action Node:** The summaries (new data) are passed to a "Slack Node" to post them in a specific channel.

Each node performs its task and then passes its output (data) to the input of the next connected node. You build this visually on a digital canvas.

V. Understanding Data and Expressions: Making Workflows Smart

This is a crucial concept, but we'll keep it straightforward:

- **Items:** Think of data in n8n as flowing in "items." An item is like a single record or a row in a spreadsheet. If you fetch 5 rows from Google Sheets, you'll have 5 items flowing through your workflow.
- **Fields:** Each item is made up of fields (like columns in a spreadsheet). For example, an item might have fields like name, email, and message.
- **Expressions:** To make your workflows dynamic, you often need to use data from a previous node in a current node. Expressions are how you do this.

Don't worry about memorising the syntax now! The key is to understand that expressions are placeholders that tell n8n: "Use the 'email' value from the data I received in the previous step." n8n has a helpful visual picker to create these expressions easily.

In other words:

- Items = Rows of data
 - Fields = Columns in each item
 - Expression = Code-like placeholder to access a field
- Example: `{{ $json["email"] }}` grabs the email from a previous step

VI. What If Something Goes Wrong? Basic Error Handling

Automations can sometimes encounter issues (e.g., an external service is temporarily down). n8n helps you manage this:

- **Continue on Error:** You can configure nodes so that if they fail, the rest of the workflow can still try to run (or a specific part of it).
- **Error Workflows:** You can design separate workflows that are triggered specifically when an error occurs in another workflow, perhaps to send you a notification.
- **Automatic Retries:** Some nodes can be set to automatically try their operation again a few times if they fail initially.

This makes your automations more resilient and reliable.

VII. Where Does n8n Run? Cloud vs. Self-Hosted

You have two main options for using n8n:

- [n8n.cloud](#): This is the hosted version provided by the n8n team. It's the easiest way to get started – no setup is required. You just sign up and begin building. **This is what we'll be using in our live session.**
- **Self-Hosted n8n**: For more advanced users or those with specific data privacy/control needs, you can install and run n8n on your server. This offers more flexibility but requires technical knowledge.

For beginners, [n8n.cloud](#) is the recommended starting point.

VIII. Getting Familiar with the n8n Interface (A Quick Preview)

When you log into n8n, you'll primarily interact with:

- **Workflow Editor (Canvas)**: The main visual area where you drag, drop, and connect nodes to build your automation.
- **Workflows List**: A dashboard showing all the workflows you've created.
- **Credentials Section**: Where you manage your connections to external apps (e.g., connecting your Google, Slack, or OpenAI accounts).
- **Executions Log**: A vital screen that shows a record of every time your workflows have run, indicating success or failure and providing details for troubleshooting.
- **Templates Gallery**: A library of pre-built workflow examples that you can copy and customise for common tasks (e.g., AI content generation, CRM updates). This is a great place to find inspiration!
- Use **"Test Node"** to check one step at a time
- Use **comments** and **naming conventions** to stay organised
- Understand input/output with the node's side panels
- Enable workflows by toggling them to "Active"

IX. Real-World Mini Example: AI-Powered Content Curation

Imagine you want to automatically find interesting articles on a specific topic, summarise them with AI, and save them to a Notion database each day.

1. **Schedule Trigger:** Runs daily.
2. **RSS Feed Reader Node / HTTP Request Node:** Fetches new articles from your favourite blogs or news sites.
3. **OpenAI Node (or similar AI node):** For each article, it generates a concise summary.
4. **Notion Node:** Adds the original article link and its AI-generated summary as a new entry in your Notion database.

This entire process, once set up, runs automatically, saving you time and effort.

X. Tips for a Smooth Start with n8n

- **Start Simple:** Don't try to build a massive, complex workflow on your first attempt. Automate one small task first.
- **Iterate and Improve:** Get a basic version working, then gradually add more features or complexity.
- **Name Your Nodes Clearly:** As workflows grow, descriptive names for each node become very helpful (e.g., "Fetch New Leads from Hubspot," "Summarise Lead Info with OpenAI").
- **Test Frequently:** Run your workflow (or parts of it) often as you build to catch issues early. Use the "Execute Node" button to test individual nodes.
- **Check the Execution Log:** If something isn't working as expected, the execution log is your best friend for understanding what data is flowing and where errors might be occurring.
- **Use Templates:** Explore the Templates Gallery for inspiration and to see how common tasks are accomplished.
- Create **sub-workflows** like "Send Email" or "Log to Notion"
- Let AI agents **call** these tools when needed

- Use **HTTP Request Nodes** when native integration isn't available

XI. Quick Recap: Key n8n Terms

- **Workflow:** Your complete automation process.
- **Node:** A single task or step in the workflow.
- **Trigger:** The node that starts the workflow.
- **Execution:** One complete run of a workflow.
- **Credentials/Connection:** Secure login information for connecting to external apps.
- **Integration:** n8n's built-in support for an external app.
- **Expression:** A way to dynamically use data from previous steps in current steps.

Final Words

You don't need a technical background to create sophisticated automations with n8n. It's designed to empower everyone to leverage the power of automation and AI. This pre-read is your foundational step. We're excited to guide you further in the live session!

Let's automate intelligently. See you there!